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A transport company is wanting to invest in more sustainable alternatives to Electric vehicles. The two technologies they are considering are Lithium ion battery Evs and Hydrogen fuel cell Evs. Both follow electrochemical processes to transform Chemical Energy to Electrical Energy to power engines, eliminating the use of combustion within their vehicles. They differ in their chemical processes, environmental and social implicaitons, manufacturing, and disposition.

Lithium ion (Li ion) batteries are Secondary Electrochemcal cells, meaning that they can be recharged. This happens when the species within the battery undergoes a redox reaction of which can be reversed during discharge and recharge. A lithium ion battery is comprised of an Anode, Cathode, Electrolyte (to balance charges), ion conductive separator (to avoid short circuiting), conductive electrodes, and external wire (to carry electrons through the components of the Ev).

Discharging
At the negatively charged Anode are sheets of Graphite with intercalated lithium ions.

$xLiC_{6\,(s)} \rightarrow xLi^{+}_{(aq)} + xe^{-} + xC_{6(s)}$

This is where oxidation occurs as the Li in xLiC6 looses xe^{-}, deintercalating xLi^{+} ions, which travels through the electrolyte towards the cathode; xe^{-} that cannot go through the electrolyte travels up the external wire through the components of the Ev and into the Positively charged Cathode. As oxidation continue to occur in the Anode, the continuous flow of electrons through the external wire is what generates electricity to power the Ev.

At the Postively charged Cathode, are Li containing Cobalt material that has been oxidised by oxygen to form a positive net ionic charge.
$Li_{1-x}CoO_{2\,(s)} + xLi^{+}\,(aq) + xe^{-} \rightarrow LiCoO_{2(s)}$

This is where reduction occurs as $Li_{1-x}CoO_2$ gains xe^{-} to form $LiCoO_2$.

The electrons that travel from the anode through the external wire they are attracted to the cathode as unlike charges attract. And because of the buildup of these neg charged

electronsxe$^-$, the positively charged xLi$^+$ are attracted to the cathode also, which is why they travel through the electrolyte towards the cathode. They all react together to make up $LiCoO_2$.

The overall equation for this process is $Li_{1-x}CoO_2 + xLiC_6 \rightarrow LiCoO_2 + xC_6$

Discharging is a spontaneous reaction that happens readily with a positive E^ocell value of E^ocell = 0.65V - (-3.05V) = + 3.70 V and the reverse for charging.

This discharging process continueuntil all/majority of the xLi$^+$ ions in the Anode has become de intercalated and travelled to the cathode. At this point, the Battery of the Ev is very low/null.

As I mentioned before, a lithium ion battery is rechargable. It does this by reversing this entire process where The initial cathode is now the anode as oxidation occurs here.

$LiCoO_2 \rightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$

Where $LiCoO_2$ is oxidised as it looses xe$^-$ which travels through the external wire to the new Cathode.

The initial Anode becomes the Cathode as this is where reduction occurs.

$xLi^{+}{}_{(aq)} + xe^- + xC_{6\,(s)} \rightarrow xLiC_{6\,(s)}$

xLi$^+$ is added xe$^-$ and is intercalated back into the sheets of graphite to form $xLiC_6$, charging the battery of the Ev.

The manufacturing of Li ion batteries go back to the extraction of the raw materials and the mining of minerals such as "Lithium, graphite, nickel, manganese and colbalt". This extraction process provides negative externalities of production onto third parties such as nearby communities/citizens from the destroyed environment and loud mining work. The environmental improprieties that come with this is the destruction of land and ecosystems that are home to plants and animals. Also acknowleging the Energy and tools required creates an opportunity cost for the other functionality these tools could have and if non renuable energy is used could be emmitting green house gasses to the atmosphere. In the production of Li ion batteries, these minerals become procesed to make the electrode materials which is then assembled to the final battery. We can see from this that the

environmental/social impacts Li ion batteries can have is not necessarily straight in the operation of the battery, but rather further up early on in the vertical supply chain. If the demand for li ion batteries increases, output increases which increases the incentive to mine more of these minerals.

Because Li ion batteries are comprised of some relatively valuable parts/materials, the disposal of Li ion batteries once they have reached their time capacity has the potential to be reused, repurposed, and recycled. This reussement provides the opportunity to decrease the demand for new minerals to make up new batteries as the old ones can be recycled. However there is a slight limitaition to the recycling of li ion batteries that pose as a challenge such as the the main being the chemical species inside, value, and technology available to properly conduct this in a reasonable amount of time, determining its viability.

Hydrogen fuel cell Evs continuously requre the use of readily made hydrogen (H_2) inorder to fuel the fuel cell and convert electrochemical energy to electricity under a redox process.

Reference my equation on encircled 8 on my cheatsheet.
Oxidation occurs at the negatively charged annode when hydrogen gas is oxidised as each of the H2 molecules loose $2e^-$ as it travels out through the external wire and the components of the eV towards the Cathode, and silmultaneously making up H^+ ions. that travel through the electrolyte towards the canode.

at the positively charged cathode reduction occurs when each H^+ ion gains an electron to make up water.

The operation of a hydrogen fuel cell deposits/creates water and heat as products and does not make combustion products such as CO_2, CO, or C. This is a ppositive benefit of using fuel cells as does not diminish the air quality ofits surroundings. However, It is important to acknolege that energy is required for the productionof H_2 which is a vital reactant the Hydrogen fuel continuously requires to operate. One way this could be done is through the hydrolysis of water, requiring an energy source to do so.
(checkout 10 on my cheatsheet for the eqn)

Similarly to the production of Lithium ion batteries, If the energy source is non-renuable ie.with the combustion of fossil fuels, thenthenit would result in negative effects on the environment and air quality contributing Green house hasses to the atmosphere. So, again despite there being no directeffects during the operation of the hydrogen fuel, there are implications to consider when looking at the molecules H2 in specific it requires to proceed to generate electricity. (E is required to make H2, depending on the source is the concern, which is used to produce Eelectric).

Comparing the two, Lithium ion batteries vs Hydrogen fuel cells, we can see a slight difference: Li ion batteries store chemical Energy which is transformed to electricity (Eelectrical) whilst hydrogen fuel cells also store chemical Energy and Supply electricity (Eeclectrical) but it is precisely that it requires Energy in the production process of H_2, a vital natural resource in the operating process of H_2 fuel cells. This energy can come from several different origins/meathods such as renewable energy sources, which contains low emissions; or perhaps non renewable energy sources like fossil fuels, which would have high emissions, inorder to allow the hydrolysis of water. The impact to the environment and society is derived/dependent from this porcess further up the supply chain rather than the actual production/operation of these hydrogen fuels inside the Ev, which is precisely why we must evaluate and acknowlege all steps and processes of each technology investigated, Li ion batteries and H_2 fuel cells.

The disposal of Li ion batteries and H_2 fuel cells when they reach the end of their lifetime capacity, also have some similarities and differences. Components in both H_2 fuel cells and li ion batteries have the potential to be reused, repurposed and recycled. For the H_2 fuel cell is the fuel cells stack, H_2 tank, etc. that can be salvaged and used once again, whereas in the Li ion battery, the materials and components can also be dismantled and reused such as litium, nickel, graphite, however it poses a little bit more challenging which would require different tools/techniques depending on the materials and the chemistry inside as there are different types of lithium batteries.

So, the main impacts of the lithium ion batteries to the environment and society are based on the mining of raw materials such as Li, Ni, C_6, Mn, Co etc... that make up the electrochemical cell and hence battery, which could ultimately disrupt the land/ecosystems by mining or perhaps the energy sources used to power these tools may not be renewable impacting both environment and communities around. An increase in

demand in Li ion batteries will increase production, increasing the demand for these raw materials, but considering there are ways to reuse these batteries, can offset this aswell.

The H_2 fuel cells negative benefits come from the energy uesed to produce H_2 resource used in the operation of the cell is produced in the first place. Similar to the lithim ion batteries and the energy used to power mining tools, would determine the potential benefits/consequence of choosing this technology. If unrenewable energy sources are used, will decrease the relative benefit of the use of hydrogen fuel being a 'sustainable' option as green house gasses are still posing as a negative externality of production further up in the supply chain. Where as renewable energy will mitigate this concern.

Holisticaly, both technology contain several positive benefits and negative consequences, which depends on the Energy used in the materials within the supply chain before arriving to the actual technology itself. So, my ultimate advice for the company is to investigate the energy sources and trace it all the way back from the very beginning (how the natural resources came to be) and tailor it to what works best for them. Perhaps they might want to expand their business of being a transport company attempt vertical integration to control more of their supply chain and really ensure that sustainable energy forms are being used the whole way until it is brought to the consumer.